

Introduction to Game Theory

Business Economics, Climate Change and the Environment

Simon Quinn

Module outline

LECTURE	DATE	TOPIC
1	9 September	Growth, Transformation and Transition
2	11 September	Introduction to Game Theory
3	16 September	Game Theory and Auctions
4	17 September	Game Theory and the Environment
5	23 September	Academic Papers Explained
6	25 September	Academic Papers Explained
7	30 September	Consumer Preferences
8	2 October	Institutions, Power and Inequality
9	7 October	Supply and Demand
10	9 October	Market Successes and Failures

Today's objectives

Today, we will:

1. Introduce **key concepts in game theory**;

Today's objectives

Today, we will:

1. Introduce **key concepts in game theory**;
2. Learn how to **solve** a game by finding the **Nash Equilibrium**;

Today's objectives

Today, we will:

1. Introduce **key concepts in game theory**;
2. Learn how to **solve** a game by finding the **Nash Equilibrium**;
3. Learn how to solve an **extensive-form game** by finding the **Subgame Perfect Equilibrium**.

Today's objectives

Today, we will:

1. Introduce **key concepts in game theory**;
2. Learn how to **solve** a game by finding the **Nash Equilibrium**;
3. Learn how to solve an **extensive-form game** by finding the **Subgame Perfect Equilibrium**.

These concepts will be important for several of the following lectures – in which we will then discuss:

Today's objectives

Today, we will:

1. Introduce **key concepts in game theory**;
2. Learn how to **solve** a game by finding the **Nash Equilibrium**;
3. Learn how to solve an **extensive-form game** by finding the **Subgame Perfect Equilibrium**.

These concepts will be important for several of the following lectures – in which we will then discuss:

1. Applications of game-theoretic ideas to **auctions** (**Lectures 3 and 4**); and

Today's objectives

Today, we will:

1. Introduce **key concepts in game theory**;
2. Learn how to **solve** a game by finding the **Nash Equilibrium**;
3. Learn how to solve an **extensive-form game** by finding the **Subgame Perfect Equilibrium**.

These concepts will be important for several of the following lectures – in which we will then discuss:

1. Applications of game-theoretic ideas to **auctions** (**Lectures 3 and 4**); and
2. Applications of game-theoretic ideas to issues of **inequality** (**Lecture 8**).

Game theory and a game show...

<https://www.youtube.com/watch?v=S0qjK3TWZE8>

On models and reality...

On models and reality...

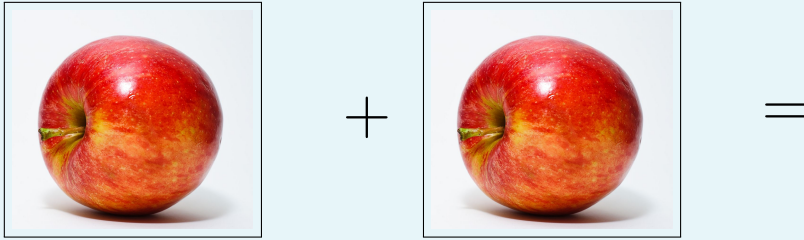


+



=

On models and reality...



Key concept: Games as models of strategic interactions

A **game** is a **formal model** of strategic interactions, which may provide useful predictions and intuition about real-world strategic behaviour.

Introducing Nick and Ibrahim...

		IBRAHIM			
		Split		Steal	
NICK	Split	£ 6800,	£ 6800	£0,	£13600
	Steal	£13600,	£0	£0,	£0

Two descriptions of 'a game'

A game, to a game theorist, is **any activity** in which your prediction of what another person will do affects what you decide to do. Such games include poker, nuclear war, love or bidding for thin air [*i.e.* 3G spectrum] in an auction.

Tim Harford, *The Undercover Economist*

Two descriptions of 'a game'

A game, to a game theorist, is **any activity** in which your prediction of what another person will do affects what you decide to do. Such games include poker, nuclear war, love or bidding for thin air [*i.e.* 3G spectrum] in an auction.

Tim Harford, *The Undercover Economist*

A game is a **description of strategic interaction** that includes the constraints on the actions that the players *can* take and the players' interests, but does not specify the actions that the players *do* take. A *solution* is a systematic description of the outcomes that may emerge in a family of games. Game theory suggests reasonable solutions for classes of games and examines their properties.

Osborne and Rubinstein (1994)

'Rice or Cassava' as a game

		Bala	
		Rice	Cassava
Anil	Rice	Both produce rice: there is a glut of rice (low price) There is a shortage of cassava Anil not producing cassava, which he is better able to produce	No market glut High prices for both crops Both farmers producing the crop for which they are less suited
	Cassava	No market glut High prices for both crops Both farmers producing the crop for which they are better suited	Both produce cassava: there is a glut of cassava (low price) There is a shortage of rice Bala not producing rice, which he is better able to produce

'Rice or Cassava' as a game in normal form

		BALA	
		Rice	Cassava
ANIL	Rice	1, 3	2, 2
	Cassava	4, 4	3, 1

'Rice or Cassava' as a game in normal form

Key questions for **defining** a game:

		BALA	
		Rice	Cassava
ANIL	Rice	1, 3	2, 2
	Cassava	4, 4	3, 1

'Rice or Cassava' as a game in normal form

Key questions for **defining** a game:

- ▶ Who are the **players**?

		BALA	
		Rice	Cassava
ANIL	Rice	1, 3	2, 2
	Cassava	4, 4	3, 1

'Rice or Cassava' as a game in normal form

Key questions for **defining** a game:

- ▶ Who are the **players**?

		BALA	
		Rice	Cassava
ANIL	Rice	1, 3	2, 2
	Cassava	4, 4	3, 1

'Rice or Cassava' as a game in normal form

		BALA	
		Rice	Cassava
ANIL	Rice	1, 3	2, 2
	Cassava	4, 4	3, 1

Key questions for **defining** a game:

- ▶ Who are the **players**?
- ▶ What **actions** can they take, and **when**?

'Rice or Cassava' as a game in normal form

		BALA	
		Rice	Cassava
ANIL	Rice	1, 3	2, 2
	Cassava	4, 4	3, 1

Key questions for **defining** a game:

- ▶ Who are the **players**?
- ▶ What **actions** can they take, and **when**?

'Rice or Cassava' as a game in normal form

		BALA	
		Rice	Cassava
ANIL	Rice	1, 3	2, 2
	Cassava	4, 4	3, 1

Key questions for **defining** a game:

- ▶ Who are the **players**?
- ▶ What **actions** can they take, and **when**?
- ▶ What **payoff** does each player get for all **combinations** of of player actions?

'Rice or Cassava' as a game in normal form

		BALA	
		Rice	Cassava
ANIL	Rice	1, 3	2, 2
	Cassava	4, 4	3, 1

Key questions for **defining** a game:

- ▶ Who are the **players**?
- ▶ What **actions** can they take, and **when**?
- ▶ What **payoff** does each player get for all **combinations** of of player actions?

'Rice or Cassava' as a game in normal form

		BALA	
		Rice	Cassava
ANIL	Rice	1, 3	2, 2
	Cassava	4, 4	3, 1

Key questions for **defining** a game:

- ▶ Who are the **players**?
- ▶ What **actions** can they take, and **when**?
- ▶ What **payoff** does each player get for all **combinations** of of player actions?
- ▶ What is the **solution concept**?

'Rice or Cassava' as a game in normal form

		BALA	
		Rice	Cassava
ANIL	Rice	1, 3	2, 2
Cassava	Cassava	4, 4	3, 1

Key questions for **defining** a game:

- ▶ Who are the **players**?
- ▶ What **actions** can they take, and **when**?
- ▶ What **payoff** does each player get for all **combinations** of of player actions?
- ▶ What is the **solution concept**?
- ▶ What are the **solutions**?

What is the **solution concept**?

Definition: Nash Equilibrium

An outcome is a Nash equilibrium if and only if **no player can deviate to another action to improve its payoff**, holding fixed all of the other players' actions. That is, a Nash equilibrium is an **intersection of best-response functions**.

Finding the Nash Equilibrium in the 'Rice-Cassava' game

		BALA	
		Rice	Cassava
ANIL	Rice	1, 3	2, 2
	Cassava	4, 4	3, 1

Finding the Nash Equilibrium in the 'Rice-Cassava' game

Questions to find the Nash Equilibrium:

		BALA	
		Rice	Cassava
ANIL	Rice	1, 3	2, 2
	Cassava	4, 4	3, 1

Finding the Nash Equilibrium in the 'Rice-Cassava' game

Questions to find the Nash Equilibrium:

- ▶ What is the best response for Anil if Bala chooses Rice?

		BALA	
		Rice	Cassava
ANIL	Rice	1, 3	2, 2
	Cassava	4, 4	3, 1

Finding the Nash Equilibrium in the 'Rice-Cassava' game

Questions to find the Nash Equilibrium:

- ▶ What is the best response for Anil if Bala chooses Rice?

		BALA	
ANIL		Rice	Cassava
Rice		1, 3	2, 2
Cassava		4, 4	3, 1

Finding the Nash Equilibrium in the 'Rice-Cassava' game

Questions to find the Nash Equilibrium:

- ▶ What is the best response for Anil if Bala chooses Rice?
- ▶ What is the best response for Anil if Bala chooses Cassava?

		BALA	
ANIL		Rice	Cassava
Rice	1,	3	2, 2
Cassava	4,	4	3, 1

Finding the Nash Equilibrium in the 'Rice-Cassava' game

Questions to find the Nash Equilibrium:

- ▶ What is the best response for Anil if Bala chooses Rice?
- ▶ What is the best response for Anil if Bala chooses Cassava?

		BALA	
ANIL		Rice	Cassava
Rice		1, 3	2, 2
Cassava		4, 4	3, 1

Finding the Nash Equilibrium in the 'Rice-Cassava' game

Questions to find the Nash Equilibrium:

- ▶ What is the best response for Anil if Bala chooses Rice?
- ▶ What is the best response for Anil if Bala chooses Cassava?
- ▶ What is the best response for Bala if Anil chooses Rice?

		BALA	
		Rice	Cassava
ANIL	Rice	1, 3	2, 2
Cassava	4, 4	3, 1	

Finding the Nash Equilibrium in the 'Rice-Cassava' game

Questions to find the Nash Equilibrium:

- ▶ What is the best response for Anil if Bala chooses Rice?
- ▶ What is the best response for Anil if Bala chooses Cassava?
- ▶ What is the best response for Bala if Anil chooses Rice?

		BALA	
		Rice	Cassava
ANIL			
Rice		1, 3	2, 2
Cassava		4, 4	3, 1

Finding the Nash Equilibrium in the 'Rice-Cassava' game

Questions to find the Nash Equilibrium:

		BALA	
ANIL		Rice	Cassava
Rice		1, 3	2, 2
Cassava		4, 4	3, 1

- ▶ What is the best response for Anil if Bala chooses Rice?
- ▶ What is the best response for Anil if Bala chooses Cassava?
- ▶ What is the best response for Bala if Anil chooses Rice?
- ▶ What is the best response for Bala if Anil chooses Cassava?

Finding the Nash Equilibrium in the 'Rice-Cassava' game

Questions to find the Nash Equilibrium:

		BALA	
ANIL		Rice	Cassava
Rice		1, 3	2, 2
Cassava		4, 4	3, 1

- ▶ What is the best response for Anil if Bala chooses Rice?
- ▶ What is the best response for Anil if Bala chooses Cassava?
- ▶ What is the best response for Bala if Anil chooses Rice?
- ▶ What is the best response for Bala if Anil chooses Cassava?

Finding the Nash Equilibrium in the 'Rice-Cassava' game

Questions to find the Nash Equilibrium:

		BALA	
		Rice	Cassava
ANIL			
Rice		1, 3	2, 2
Cassava		4, 4	3, 1

- ▶ What is the best response for Anil if Bala chooses Rice?
- ▶ What is the best response for Anil if Bala chooses Cassava?
- ▶ What is the best response for Bala if Anil chooses Rice?
- ▶ What is the best response for Bala if Anil chooses Cassava?
- ▶ What is the Nash Equilibrium?

Finding the Nash Equilibrium in the 'Rice-Cassava' game

Questions to find the Nash Equilibrium:

		BALA	
ANIL		Rice	Cassava
Rice		1, 3	2, 2
Cassava		4, 4	3, 1

- ▶ What is the best response for Anil if Bala chooses Rice?
- ▶ What is the best response for Anil if Bala chooses Cassava?
- ▶ What is the best response for Bala if Anil chooses Rice?
- ▶ What is the best response for Bala if Anil chooses Cassava?
- ▶ What is the Nash Equilibrium?

'When self-interest works: The invisible hand'

		Bala	
		Rice	Cassava
Anil	Rice	Both produce rice: there is a glut of rice (low price) There is a shortage of cassava Anil not producing cassava, which he is better able to produce	No market glut High prices for both crops Both farmers producing the crop for which they are less suited
	Cassava	No market glut High prices for both crops Both farmers producing the crop for which they are better suited	Both produce cassava: there is a glut of cassava (low price) There is a shortage of rice Bala not producing rice, which he is better able to produce

'When self-interest works: The invisible hand'

		Bala	
		Rice	Cassava
Anil	Rice	Both produce rice: there is a glut of rice (low price) There is a shortage of cassava Anil not producing cassava, which he is better able to produce	No market glut High prices for both crops Both farmers producing the crop for which they are less suited
	Cassava	No market glut High prices for both crops Both farmers producing the crop for which they are better suited	Both produce cassava: there is a glut of cassava (low price) There is a shortage of rice Bala not producing rice, which he is better able to produce

Pest control as a game

		Bala	
		IPC	Terminator
Anil	IPC	Beneficial insects spread over both fields, eliminating pests No water contamination	Bala's chemicals spread to Anil's field and kill his beneficial insects Limited water contamination
	Terminator	Anil's chemicals spread to Bala's field and kill his beneficial insects Limited water contamination	Eliminates all pests Heavy water contamination Requires costly filtration system

Finding the Nash Equilibrium in the pest-control game

		BALA	
		IPC	Terminator
ANIL	IPC	3, 3	1, 4
Terminator	IPC	4, 1	2, 2

Finding the Nash Equilibrium in the pest-control game

Questions to find the Nash:

		BALA	
ANIL		IPC	Terminator
IPC	IPC	3, 3	1, 4
	Terminator	4, 1	2, 2

Finding the Nash Equilibrium in the pest-control game

Questions to find the Nash:

- ▶ What is the best response for Anil if Bala chooses IPC?

		BALA	
ANIL		IPC	Terminator
IPC	3,	3	1, 4
Terminator	4,	1	2, 2

Finding the Nash Equilibrium in the pest-control game

Questions to find the Nash:

- ▶ What is the best response for Anil if Bala chooses IPC?

		BALA	
		IPC	Terminator
ANIL	IPC	3, 3	1, 4
Terminator	4, 1	2, 2	

Finding the Nash Equilibrium in the pest-control game

Questions to find the Nash:

- ▶ What is the best response for Anil if Bala chooses IPC?
- ▶ What is the best response for Anil if Bala chooses Terminator?

		BALA	
		IPC	Terminator
ANIL	IPC	3, 3	1, 4
Terminator	IPC	4, 1	2, 2

Finding the Nash Equilibrium in the pest-control game

Questions to find the Nash:

- ▶ What is the best response for Anil if Bala chooses IPC?
- ▶ What is the best response for Anil if Bala chooses Terminator?

		BALA	
ANIL		IPC	Terminator
IPC	3,	3	1, 4
Terminator	4,	1	2, 2

Finding the Nash Equilibrium in the pest-control game

		BALA	
ANIL		IPC	Terminator
IPC		3, 3	1, 4
Terminator		4, 1	2, 2

Questions to find the Nash:

- ▶ What is the best response for Anil if Bala chooses IPC?
- ▶ What is the best response for Anil if Bala chooses Terminator?
- ▶ What is the best response for Bala if Anil chooses IPC?

Finding the Nash Equilibrium in the pest-control game

		BALA	
ANIL		IPC	Terminator
IPC	3,	3	1, 4
Terminator	4,	1	2, 2

Questions to find the Nash:

- ▶ What is the best response for Anil if Bala chooses IPC?
- ▶ What is the best response for Anil if Bala chooses Terminator?
- ▶ What is the best response for Bala if Anil chooses IPC?

Finding the Nash Equilibrium in the pest-control game

		BALA	
ANIL		IPC	Terminator
IPC	3,	3	1, 4
Terminator	4,	1	2, 2

Questions to find the Nash:

- ▶ What is the best response for Anil if Bala chooses IPC?
- ▶ What is the best response for Anil if Bala chooses Terminator?
- ▶ What is the best response for Bala if Anil chooses IPC?
- ▶ What is the best response for Bala if Anil chooses Terminator?

Finding the Nash Equilibrium in the pest-control game

		BALA	
		IPC	Terminator
ANIL	IPC	3, 3	1, 4
	Terminator	4, 1	2, 2

Questions to find the Nash:

- ▶ What is the best response for Anil if Bala chooses IPC?
- ▶ What is the best response for Anil if Bala chooses Terminator?
- ▶ What is the best response for Bala if Anil chooses IPC?
- ▶ What is the best response for Bala if Anil chooses Terminator?

Finding the Nash Equilibrium in the pest-control game

		BALA	
ANIL		IPC	Terminator
IPC	3,	3	1, 4
Terminator	4,	1	2, 2

Questions to find the Nash:

- ▶ What is the best response for Anil if Bala chooses IPC?
- ▶ What is the best response for Anil if Bala chooses Terminator?
- ▶ What is the best response for Bala if Anil chooses IPC?
- ▶ What is the best response for Bala if Anil chooses Terminator?
- ▶ What is the Nash Equilibrium?

Finding the Nash Equilibrium in the pest-control game

		BALA	
		IPC	Terminator
ANIL	IPC	3, 3	1, 4
Terminator	4, 1	2, 2	

Questions to find the Nash:

- ▶ What is the best response for Anil if Bala chooses IPC?
- ▶ What is the best response for Anil if Bala chooses Terminator?
- ▶ What is the best response for Bala if Anil chooses IPC?
- ▶ What is the best response for Bala if Anil chooses Terminator?
- ▶ What is the Nash Equilibrium?

'When self-interest doesn't work: The prisoners' dilemma'

		Bala	
		IPC	Terminator
Anil	IPC	Beneficial insects spread over both fields, eliminating pests No water contamination	Bala's chemicals spread to Anil's field and kill his beneficial insects Limited water contamination
	Terminator	Anil's chemicals spread to Bala's field and kill his beneficial insects Limited water contamination	Eliminates all pests Heavy water contamination Requires costly filtration system

'When self-interest doesn't work: The prisoners' dilemma'

		Bala	
		IPC	Terminator
Anil	IPC	Beneficial insects spread over both fields, eliminating pests No water contamination	Bala's chemicals spread to Anil's field and kill his beneficial insects Limited water contamination
	Terminator	Anil's chemicals spread to Bala's field and kill his beneficial insects Limited water contamination	Eliminates all pests Heavy water contamination Requires costly filtration system

On the role of self-interest

Key concept: Self-interest and optimal outcomes

If each player pursues their self-interest, this **may** lead to the optimal overall outcome. But it also **may not** – this depends upon the structure of the game.

On the role of self-interest

Key concept: Self-interest and optimal outcomes

If each player pursues their self-interest, this **may** lead to the optimal overall outcome. But it also **may not** – this depends upon the structure of the game.

It is not from the benevolence of the butcher, the brewer, or the baker that we expect our dinner, but from their regard to their own self-interest.

Adam Smith, *The Wealth of Nations* (1776)

Back to Nick and Ibrahim...

		IBRAHIM			
		Split		Steal	
NICK	Split	£ 6800,	£ 6800	£0,	£13600
	Steal	£13600,	£0	£0,	£0

Back to Nick and Ibrahim...

		IBRAHIM			
		Split		Steal	
NICK	Split	£ 6800,	£ 6800	£0,	£13600
	Steal	£13600,	£0	£0,	£0

Over to you...

Can you find the **Nash Equilibrium** (perhaps 'Nash Equilibria') of this game?

Back to Nick and Ibrahim...

		IBRAHIM			
		Split		Steal	
NICK	Split	£ 6800,	£ 6800	£0,	£13600
	Steal	£13600,	£0	£0,	£0

Over to you...

Can you find the **Nash Equilibrium** (perhaps 'Nash Equilibria') of this game?

<https://www.youtube.com/watch?v=S0qjK3TWZE8>

<https://radiolab.org/episodes/golden-rule>

<https://radiolab.org/episodes/104010-one-good-deed-deserves-another>

Definition: Strict domination

For Player 1 (for example), action a **strictly dominates** action b if Player 1 receives a strictly higher payoff from a than from b , irrespective of Player 2's actions. In that situation, we would say that action b is **strictly dominated**.

A hypothetical conversation with a friend...

		BALA	
		IPC	Terminator
ANIL	IPC	3, 3	1, 4
Terminator	IPC	4, 1	2, 2

A hypothetical conversation with a friend...

		BALA	
		IPC	Terminator
ANIL	IPC	3, 3	1, 4
Terminator	IPC	4, 1	2, 2

“Which action would you choose, and why?”

A hypothetical conversation with a friend...

		BALA	
		IPC	Terminator
ANIL	IPC	3, 3	1, 4
Terminator	IPC	4, 1	2, 2

“Which action would you choose, and why?”

“I would choose IPC. Because, if I choose IPC, the other player would do the same. And the payoff of 3 is better than the payoff of 2.”

A hypothetical conversation with a friend...

		BALA	
ANIL		IPC	Terminator
IPC	3,	3	1, 4
Terminator	4,	1	2, 2

“Which action would you choose, and why?”

“I would choose IPC. Because, if I choose IPC, the other player would do the same. And the payoff of 3 is better than the payoff of 2.”

Definition: Obviously Dominant strategy

For Player 1 (for example), action a is **obviously dominant** if, at the time when the player must choose, the best possible outcome from any other action is no better than the worst possible outcome from action a .

A hypothetical conversation with a friend...

		BALA	
		IPC	Terminator
ANIL	IPC	3, 3	1, 4
Terminator	IPC	104, 1	102, 2

A hypothetical conversation with a friend...

		BALA	
		IPC	Terminator
ANIL	IPC	3, 3	1, 4
	Terminator	104, 1	102, 2

“Which action would you choose, and why?”

“I would choose Terminator. Because the **worst** outcome if I choose Terminator is better than the **best** outcome if I choose IPC.”

A hypothetical conversation with a friend...

		BALA	
		IPC	Terminator
ANIL	IPC	3, 3	1, 4
Terminator	104, 1	102, 2	

“Which action would you choose, and why?”

“I would choose Terminator. Because the **worst** outcome if I choose Terminator is better than the **best** outcome if I choose IPC.”

Climate policy as a 'Hawk-Dove' game

		US		
		Restrict	BAU	
China	Restrict	Reduction in emissions sufficient to moderate climate change	US free rides on Chinese emissions cutbacks	← Temperatures continue to rise, imposing large but bearable costs
	BAU	China free rides on US emissions cutbacks	No reduction in emissions	← Catastrophic, irreversible climate change ← Temperatures continue to rise, imposing large but bearable costs

A 'Hawk-Dove' game: Can you find the Nash Equilibrium?

		USA	
		Restrict	BAU
CHINA	Restrict	10, 10	4, 12
	BAU	12, 4	0, 0

A 'Hawk-Dove' game: Can you find the Nash Equilibrium?

		USA	
		Restrict	BAU
CHINA	Restrict	10, 10	4, 12
	BAU	12, 4	0, 0

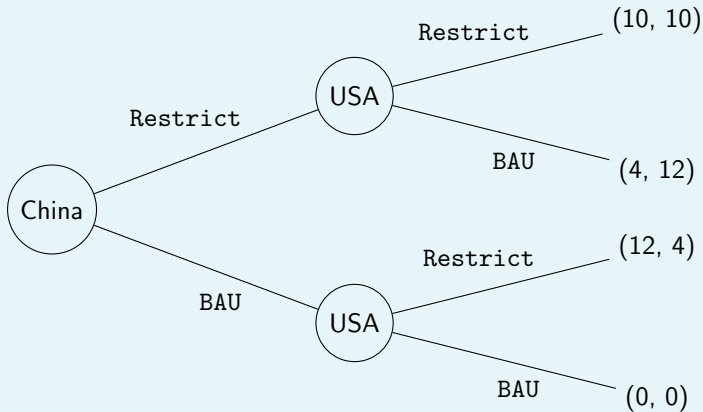
A 'Hawk-Dove' game: Can you find the Nash Equilibrium?

		USA	
		Restrict	BAU
CHINA	Restrict	10, 10	4, 12
	BAU	12, 4	0, 0

Key concept: Multiple equilibria

Some games can have **more than one** Nash Equilibrium (as we saw earlier).

A game in 'extensive form'



Definition: Subgame

For our purposes, a **subgame** is a game that begins at a node in an extensive form game.

Definition: Subgame

For our purposes, a **subgame** is a game that begins at a node in an extensive form game.

Definition: Subgame Perfect Equilibrium

A **subgame perfect equilibrium** is a combination of strategies that induces a Nash equilibrium in every subgame.

Definition: Subgame

For our purposes, a **subgame** is a game that begins at a node in an extensive form game.

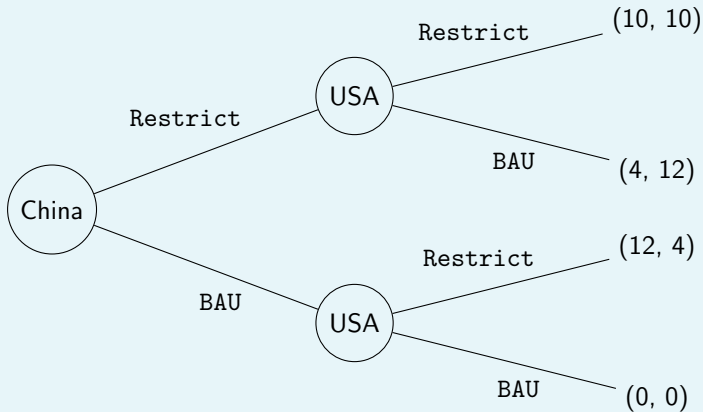
Definition: Subgame Perfect Equilibrium

A **subgame perfect equilibrium** is a combination of strategies that induces a Nash equilibrium in every subgame.

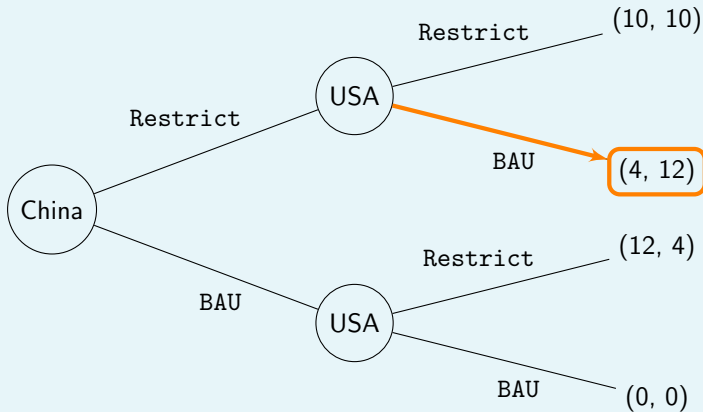
Key concept: Backward induction

We can find the Subgame Perfect Equilibrium by using '**backward induction**' from the final period.

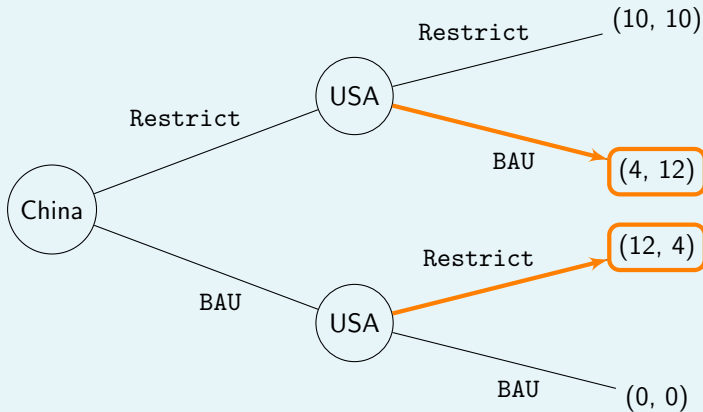
Applying backward induction



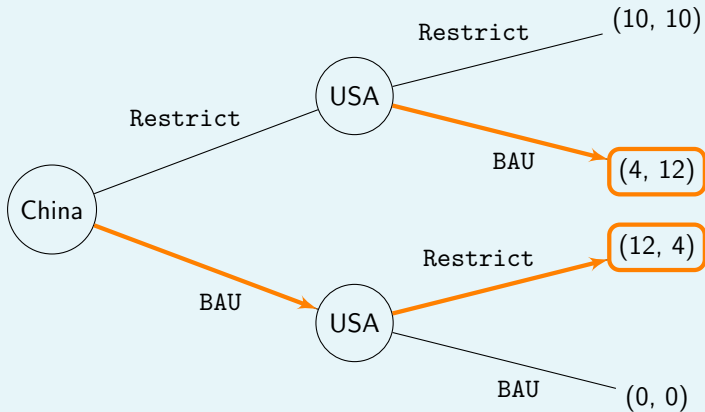
Applying backward induction



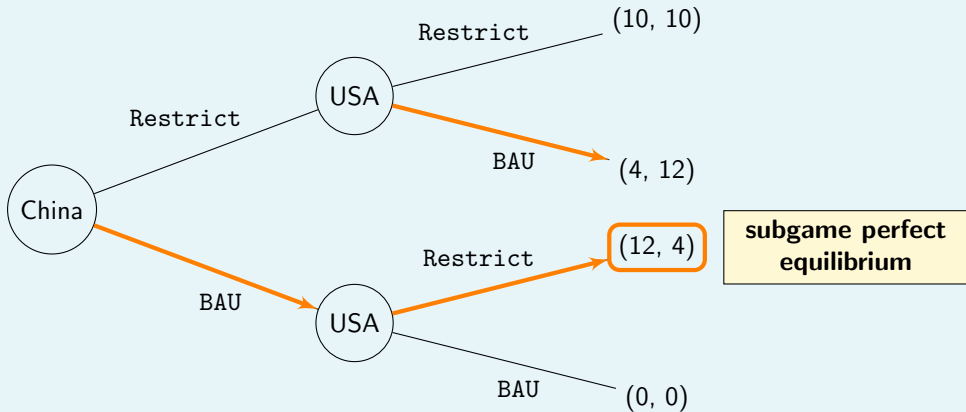
Applying backward induction



Applying backward induction



Applying backward induction



Initiative and the 'First-Mover Advantage'

Key concept: First-mover Advantage

In some games, there is an advantage to moving first: a '**first-mover advantage**'. This describes strategic interactions where it is useful to take the initiative.

Initiative and the 'First-Mover Advantage'

Key concept: First-mover Advantage

In some games, there is an advantage to moving first: a '**first-mover advantage**'. This describes strategic interactions where it is useful to take the initiative.

He that plays first, is understood to have the attack.

Joseph Bertin, *The Noble Game of Chess* (1735)

Today's objectives

Today, we have:

1. Introduced **key concepts in game theory**;

Today's objectives

Today, we have:

1. Introduced **key concepts in game theory**;
2. Learned how to **solve** a game by finding the **Nash Equilibrium**;

Today's objectives

Today, we have:

1. Introduced **key concepts in game theory**;
2. Learned how to **solve** a game by finding the **Nash Equilibrium**;
3. Learned how to solve an **extensive-form game** by finding the **Subgame Perfect Equilibrium**.

Today's objectives

Today, we have:

1. Introduced **key concepts in game theory**;
2. Learned how to **solve** a game by finding the **Nash Equilibrium**;
3. Learned how to solve an **extensive-form game** by finding the **Subgame Perfect Equilibrium**.

Today's objectives

Today, we have:

1. Introduced **key concepts in game theory**;
 2. Learned how to **solve** a game by finding the **Nash Equilibrium**;
 3. Learned how to solve an **extensive-form game** by finding the **Subgame Perfect Equilibrium**.
-
1. Applications of game-theoretic ideas to **auctions** (**Lectures 3 and 4**); and

Today's objectives

Today, we have:

1. Introduced **key concepts in game theory**;
 2. Learned how to **solve** a game by finding the **Nash Equilibrium**;
 3. Learned how to solve an **extensive-form game** by finding the **Subgame Perfect Equilibrium**.
-
1. Applications of game-theoretic ideas to **auctions** (**Lectures 3 and 4**); and
 2. Applications of game-theoretic ideas to issues of **inequality** (**Lecture 8**).

Introduction to Game Theory

Business Economics, Climate Change and the Environment

Simon Quinn